



BlockSign



CUSTOMER DELIVERY DOCUMENT · E-SIGN

Technical Product Description

Customer-oriented functional and technical description of BlockSign E-Sign, including account access, profile and team administration, multilingual workflows, document preparation, electronic signature levels, Swisscom Trust Services integration, evidence, security and legal classification.

VERSION

V1.0

DATE

4 September 2026

STATUS

Technical product profile

LANGUAGE

English

Document boundary

This document explains the current product model and intended customer workflow. It is not legal advice, a service-level agreement, a certificate, or a guarantee that a particular electronic signature satisfies every statutory form requirement.

START HERE

1. Quickstart — send a document for signature

From account creation to the completed PDF in six clear steps. BlockSign prepares the uploaded document automatically in the background.



1

Sign up

Create your BlockSign account or sign in with an enabled login provider.



2

Upload

Select the PDF you want to send for electronic signature.



3

Invite

Add the signees, choose their signature level and send the invitations.



4

Pay

Complete payment when the selected document or signature level requires it.



5

Collect signatures

Wait for the invited signees and, when needed, add your own signature.

Wait for
signees

Sign yourself
· optional



6

Download

Download the completed signed PDF and the available evidence.



Automatic background preparation

After upload, BlockSign processes the PDF, creates the preview and prepares the signing workspace while you continue with the invitation flow.

Practical first test

Start with a non-sensitive sample PDF and one internal recipient. Confirm invitation delivery, recipient access, field placement, signature completion and final download before rolling the workflow out to external customers.

PRODUCT SCOPE

2. Platform overview

BlockSign E-Sign is a tenant-aware electronic signature platform that combines document workflow management with configurable signature levels and external trust-service integration. The web application coordinates the process; regulated FES/QES identity and signature functions are performed through the configured trust service provider.

Document workspace

Upload, processing, preview, recipients, field preparation, signing order, reminders, status views and authorised download.

Identity and access

Account registration, login, enabled OAuth providers, email-bound invitations, profile settings, optional factors and provider enrolment.

Evidence and delivery

Signature output, timestamps and provider evidence where applicable, document hashes, workflow communication, reporting and integrity verification.

High-level trust architecture

Sender & signer browsers

Authenticated web sessions, invitation links and explicit user actions



BlockSign E-Sign

Workflow, access, documents, billing entitlements, audit context and provider orchestration



Swisscom Trust Services

Provider-side FES/QES identity, credential and cryptographic signature process

IDENTITY AND ACCESS

3. Sign-up, login and connected identity providers

Sign-up

Registration captures the user name, unique email address and password confirmation. A company can be created during registration when company information is supplied. The account is signed in and a signup email is queued.

Password login

Existing users authenticate with their registered credentials. Password reset and logout paths are available. Access remains subject to account state and the authorisation rules of the requested resource.

OAuth login

Google, Microsoft and Apple OAuth entry points are implemented. These should be described as enabled connected-login options, not as arbitrary enterprise SAML/OIDC support unless separately contracted and configured.

Invitation identity

A signature invitation is associated with the recipient address supplied by the sender. Recipient access checks use that identity context; possession of an unrelated signed-in account does not grant access to another recipient's document.

API access

Authenticated users can manage an API key from the profile area. API requests are protected by bearer-key middleware and remain subject to endpoint-level permissions and resource ownership checks. API credentials must be stored as secrets and must never be embedded in client-side code, emails or documents.

USER ADMINISTRATION

4. Profile management

The profile area acts as the user's control surface for identity, preferences, security and signature readiness.

AREA	PURPOSE	IMPORTANT BOUNDARY
Personal profile	Maintain account-facing profile information and view the identity attached to the account.	Read-only contexts must not silently mutate identity data.
Language preference	Persist the selected supported locale for future authenticated sessions.	Unsupported locale values are rejected or replaced by the configured fallback.
Security factors	Manage available MFA/biometric factors and phone-related verification steps.	Biometrics and phone verification are authentication factors; they are not themselves a legal FES/QES level.
Signature readiness	View and initiate legal-framework-specific FES or QES upgrade and onboarding paths.	A payment entitlement never replaces successful provider identity evidence.
Company profile	Maintain company details and company branding where the account is authorised to do so.	Company operations remain scoped to company membership and role.
API key	Generate or manage the credential used by supported customer API calls.	Rotation and secure storage are operational customer responsibilities.

ORGANISATION COLLABORATION

5. Team management and invitation emails

Company administrators can invite users to join a company team. Team membership supports organisational collaboration while preserving user-level authentication and document-level authorisation.

1

Administrator starts an invitation

The target email address is checked against existing members and pending invitations.

2

BlockSign creates a one-time invitation context

The invitation uses a high-entropy token and an expiry window. The current implementation uses a seven-day expiry for team invitations.

3

Branded email is queued

Sender identity, tenant branding, locale and callback URL are captured for asynchronous delivery.

4**Recipient accepts**

The server validates the invitation and associates the user with the intended company according to the supported membership model.

Admin

Administrative company and team capabilities, subject to application checks.

Member

Normal company membership with access determined by feature and document authorisation.

Disabled

Membership state used to prevent normal active access.

Email delivery dependency

BlockSign queues the message and maintains invitation state. Final delivery also depends on the configured mail provider, recipient server and anti-spam controls; it must not be treated as guaranteed delivery.

INTERNATIONAL USE

6. Multilingual operation

The current product catalogue provides German, English, French and Spanish application content. Locale selection is validated against the configured language list.

DE**Deutsch**

German user interface and transactional content where translated.

EN**English**

English user interface and transactional content where translated.

FR

Français

French user interface and transactional content where translated.

ES

Español

Spanish user interface and transactional content where translated.

Locale resolution

1. A valid language selected in the current session takes precedence.
2. Otherwise, a valid locale stored on the authenticated user profile is used.
3. If neither is available, the application uses the configured fallback locale.
4. Queued emails capture locale and tenant context when dispatched so that background workers do not depend on a later browser session.

Translation scope

The application maintains catalogue parity checks for supported languages. Customer-specific copy, legacy templates and newly introduced messages still require release review as part of localisation quality assurance.

7. AES, FES and QES

LEVEL	PRODUCT INTERPRETATION	IDENTITY AND PROVIDER MODEL	TYPICAL LEGAL POSITIONING
AES STANDARD	Simple electronic signature for workflows that do not require a regulated higher level.	In-house synchronous process associated with the authenticated or email-bound recipient context. A verified email address or phone number can support attribution.	Electronic evidence for many everyday transactions; it does not automatically satisfy a statutory written-form requirement.
FES PROVIDER-BACKED	Advanced electronic signature with stronger signer attribution and integrity controls.	External trust-provider process. BlockSign requires the configured provider profile, verified attributes and successful legal-framework-specific evidence.	Higher evidentiary value than AES. It is not automatically equivalent to handwriting where qualified written form is required.
QES QUALIFIED	Qualified electronic signature created through the configured qualified trust-service process.	Requires provider-backed qualified identification, credential and explicit authorisation for the concrete signature request.	Highest regulated signature level and generally equivalent to a handwritten signature within the applicable framework, subject to statutory exceptions.

Authentication is not a signature level

Biometrics, OTP and phone verification can strengthen authentication. They do not independently convert an AES workflow into FES or QES and do not replace a Swisscom provider record.

Legal-framework profiles

eIDAS

European trust-services framework used for configured EU-oriented FES and QES profiles. Signers with a supported EU passport can use the eIDAS onboarding path. Suitability still depends on the transaction and applicable law.

ZertES

Swiss framework used for configured Switzerland-oriented FES and QES profiles. Signers with a Swiss passport use the ZertES onboarding path. It is maintained as a separate entitlement and evidence scope.

Separation rule: an eIDAS level does not automatically grant the corresponding ZertES level, and vice versa. Existing provider evidence reduces upgrade requirements only in the exactly matching legal framework.

Identity document is not governing law

The Swiss-passport and supported EU-passport paths describe provider onboarding. The law governing the document, the parties, the transaction type and any authority-specific instructions still determine the required signature form.

Practical legal guideline: choosing AES, FES or QES

This guide is a risk- and form-based starting point, not legal advice. Always check the governing law, contractual form clause, regulated-sector rules and the receiving authority's instructions. Where requirements conflict or remain unclear, choose QES or obtain jurisdiction-specific legal advice before signing.

CHOOSE	SUITABLE STARTING POINT	DO NOT RELY ON IT WHEN	TYPICAL EXAMPLES
AES	No statutory or agreed written-form requirement applies; transaction risk is ordinary; email/account attribution and the audit trail provide proportionate evidence.	A law, contract, regulator, court, registry or tender expressly requires an advanced, qualified, handwritten, notarised or witnessed form.	Routine approvals, acknowledgements, standard B2B agreements, many NDAs and ordinary U.S. commercial contracts.
FES	Stronger identity binding, sole control and tamper evidence are commercially or evidentially important, but no rule requires QES or another special form.	The applicable rule requires the electronic equivalent of handwritten signature, explicitly names QES, or excludes electronic form.	Higher-value commercial agreements, controlled internal approvals, regulated workflows that accept advanced signatures, and transactions where attribution may be disputed.
QES	The electronic process must provide the eIDAS equivalent of a handwritten signature, a statutory written-form requirement may be replaced electronically, or the receiving authority/tender expressly requires QES.	The law requires notarisation, witnesses, a specific registry procedure, an original paper instrument, or expressly excludes electronic form.	German fixed-term employment agreements, where the fixed-term clause requires written form and electronic form is used; public-tender declarations whose procurement documents require QES; other high-risk or formally regulated documents.

EU and Germany

- Under eIDAS Article 25, an electronic signature cannot be denied legal effect solely because it is electronic or non-qualified. A QES has the equivalent legal effect of a handwritten signature across the EU.
- eIDAS does not itself decide whether a particular contract is valid or subject to a mandatory form; national or sector-specific law still applies.
- In Germany, § 14(4) TzBfG requires written form for the fixed-term limitation of an employment contract. If electronic form is used, § 126a BGB requires each party to add its name and apply a QES to an equivalent document. The fact that the term is one year does not create the rule—the fixed term does.
- Some employment acts cannot be signed electronically at all. For example, § 623 BGB excludes electronic form for termination agreements and notices of termination.

Public tenders and the United States

- There is no blanket rule that every EU public tender requires QES. Follow the exact procurement documents and submission portal. When a declaration or tender instruction says that only QES is accepted, AES and FES are insufficient.
- European Commission tender specifications commonly require QES for electronically signed formal declarations. This is an authority- and procedure-specific requirement, not a universal rule for every procurement document.
- In the United States, the federal E-SIGN Act generally prevents signatures and contracts in interstate or foreign commerce from being denied validity solely because they are electronic. AES-style workflows can therefore be suitable for many ordinary contracts.
- U.S. exceptions and overlays remain important, including wills and testamentary trusts, family-law matters, specified UCC matters, court documents and certain critical notices. State UETA variations, consumer-consent rules and sector-specific law must also be checked.

Escalate before signing

Use legal review rather than a product default for employment termination, deeds or notarised acts, guarantees with special form rules, consumer-credit or insurance notices, court filings, land transactions, public registers, cross-border regulated transactions, or any tender that prescribes a signature technology.

Primary legal references

- Regulation (EU) No 910/2014 (eIDAS), especially Articles 25 and 27.
- Germany: § 14(4) TzBfG—written form for the fixed-term limitation of an employment contract.
- Germany: § 126a BGB—replacement of statutory written form by electronic form using QES.
- Germany: § 623 BGB—electronic form excluded for termination agreements and notices.
- United States: 15 U.S.C. § 7001—general validity of electronic signatures and records.
- United States: 15 U.S.C. § 7003—specific E-SIGN Act exceptions.

Legal-source review date: 4 September 2026. Laws, administrative practice and tender requirements can change; re-check the current official source before relying on this guide.

UPGRADE AND ENROLMENT

8. Paid onboarding and signature upgrade path

AES is the default entry level. FES and QES can require a paid, framework-specific onboarding entitlement followed by successful provider enrolment.

1. Quote and payment

BlockSign calculates the missing level for the selected legal framework and verifies the supported payment path.



2. Onboarding entitlement

Successful server-side settlement unlocks the relevant enrolment path; it does not create identity evidence.



3. Swisscom evidence

Only successful provider-side identification and credential evidence activates provider-backed readiness.

Upgrade rules

PATH	CURRENT PRICE PER LEGAL FRAMEWORK	RESULT AFTER PAYMENT	RESULT AFTER PROVIDER COMPLETION
AES	Free	No paid onboarding required	AES remains available under its normal attribution rules
AES → FES	€15	FES onboarding entitlement	FES readiness after valid matching provider evidence
AES → QES	€35 total	QES onboarding entitlement	QES readiness after valid matching qualified provider evidence
FES → QES	€20 difference	QES onboarding entitlement	QES readiness after valid matching provider evidence

Prices are product configuration documented at the date of this profile and may be superseded by a commercial agreement. eIDAS and ZertES are accounted for separately.

Self-service upgrade

A user can initiate an upgrade from the profile or signing journey and pays for their own missing level.

Inviter-funded upgrade

A sender can prepay the maximum required onboarding amount for an invited signer. The final legal-framework selection determines application; unused value can be recorded as credit.

Critical control

Browser return parameters are not proof of payment. BlockSign must verify provider status, amount, currency, application payer, order and provider reference server-side before granting an onboarding entitlement.

END-TO-END PROCESS

9. Complete document and signature flow

1

Upload

The sender uploads a supported document. BlockSign creates the document record, processes the source and prepares protected working artefacts.

2

Preview and preparation

Preview images and extracted content support the preparation workspace. The sender reviews the document and configures recipient fields and order.

3

AI-assisted analysis

Available AI analysis can help identify document context and preparation opportunities. Output is assistance only—not legal advice, a guaranteed clause review or an autonomous signing decision.

4

Recipient invitation

The sender enters recipient addresses. BlockSign creates recipient-specific state and queues branded invitation messages.

5

Email-bound access

The signer follows the invitation and authenticates in the expected recipient context. AES can use a protected invitation session; provider-backed FES/QES requires full account login and stronger verified attributes.

6

Eligibility decision

BlockSign resolves the required level and legal framework. A ready signer continues; another signer is directed through the paid entitlement and provider enrolment path.

7

Review and explicit signature approval

The signer reviews the final document and approves the concrete signature action. Provider authorisation is bound to the specific signing request.

8

Signature creation

AES is produced within the BlockSign workflow. For Swisscom-backed FES/QES, BlockSign submits the bound request and incorporates the returned provider signature and validation evidence.

9

Workflow progression

For multi-recipient documents, the application advances according to the configured sequence and records status changes.

10

Completion, email and download

Completion communication is queued. Authorised parties can access the completed document; download access is not broader than view access.

COMMUNICATION LIFECYCLE

10. Email and notification model

Transactional communication is queued so that document requests are not blocked by mail-provider response time. The queued context includes the relevant locale and tenant identity.

Account

- Signup communication
- Password-related communication
- Team confirmation invitation

Signature workflow

- Signer invitation
- Reminder
- Review or POD invitation
- Cancellation

Completion

- Signature confirmation
- Finished-document notice
- Invoice and payment-related messages

Tenant-aware delivery

Mail sender, brand, callback URL and language must be taken from the dispatch context—not guessed from a background worker's host or an external recipient domain.

11. Role of Swisscom Trust Services

Swisscom Trust Services is the configured external trust-service provider for personal FES and QES processes. BlockSign orchestrates the customer workflow; Swisscom performs the provider-side identity, credential and cryptographic signature functions required by the selected signing profile.

BlockSign responsibilities

- Document upload, preparation and workflow state
- Recipient invitation and application access control
- Selection of level and legal framework
- Upgrade quote, payment verification and entitlement
- Provider request orchestration and callback handling
- Integration of the returned signature into the PDF
- Evidence presentation, communication and download

Swisscom responsibilities

- Provider-side enrolment and identity checks
- Signing credential and certificate context
- Explicit authorisation of the concrete signature
- Cryptographic creation of FES/QES signatures
- Provider validation and revocation information
- Qualified trust-service evidence where applicable

No local substitution

A legacy KYC flag, biometric factor, verified phone, payment or record from another provider is not equivalent to current Swisscom FES/QES evidence. If valid provider evidence is unavailable or expired, the signer returns to the appropriate provider journey.

12. Storage, integrity and verification

CONTROL	TECHNICAL PURPOSE	CUSTOMER INTERPRETATION
Object storage	Production documents use durable object storage with document-specific key namespaces. Local files are a development/test aid rather than the production durability model.	Document access must be mediated by application authorisation; storage location alone is not an access decision.
Document hash	A final SHA-256 document hash is recorded for integrity comparison and reporting.	The stored final hash is the authoritative BlockSign integrity reference.
PDF signature inspection	Independent tooling can inspect PDF byte ranges and embedded signature results.	Byte-integrity inspection must not be presented as universal legal or certificate-chain validation.
Public verification	The verification workflow compares uploaded document evidence, hashes and embedded metadata where available.	It supports integrity verification; its exact output depends on the evidence contained in the document and platform record.
Blockchain evidence	Cardano anchoring can provide supplementary timestamped integrity evidence.	Blockchain evidence is additive. It is not required to replace the authoritative stored final document hash.

INFORMATION SECURITY

13. Security, tenant boundaries and ISO/IEC 27001

BlockSign applies layered controls across authenticated sessions, tenant-aware branding and communication, company membership, recipient-bound access, document ownership and provider callbacks.

Tenant context

Host and tenant configuration determine brand, legal identity, URLs and mail presentation. Background operations retain explicit tenant context.

Resource authorisation

Document access requires ownership or the authenticated invited-recipient relationship. Direct identifiers do not by themselves grant access.

External callbacks

Payment and trust-provider callbacks require server-side correlation and verification. Browser parameters are not authoritative evidence.

ISO/IEC 27001

BlockSign's published product material identifies ISO/IEC 27001 certification. ISO/IEC 27001 concerns the information security management system—not the legal qualification of an individual signature. Certificate number, certified scope, certification body and validity evidence should be supplied in the applicable customer due-diligence or delivery package.

Operational security expectations

- Use secrets management for API keys, signing credentials, callback secrets and mail-provider credentials.
- Apply least privilege to user roles, team administration, storage access and provider integrations.
- Maintain logging and monitoring without recording passwords, tokens, identity payloads or signing secrets.
- Keep payment settlement idempotent and bind each paid capability to one verified provider reference.
- Review access and signature eligibility again immediately before a consequential signing operation.

REGULATORY POSITIONING

14. Legal classification and customer responsibility

The required signature level depends on the applicable law, the type of transaction, contractual evidence requirements and any statutory form requirement.

AES

Often suitable for ordinary commercial workflows. Evidentiary strength depends on attribution, authentication, document integrity and the audit context.

FES

Adds stronger identity linkage and tamper detection. It does not automatically satisfy every statutory written-form requirement.

QES

Highest regulated level and generally equivalent to handwriting within its applicable framework, subject to exclusions and special formalities.

Possible exceptions

A QES does not replace requirements such as notarisation, public authentication, personal presence, authority registration or another specifically mandated form where the applicable law excludes electronic substitution.

BlockSign provides the technical workflow, access controls, evidence handling and provider integration. The customer remains responsible for selecting a suitable signature level and legal framework for each use case and should obtain legal advice where the classification is uncertain.

SERVICE OPERATION

15. Dependencies and failure behaviour

DEPENDENCY	USED FOR	EXPECTED PRODUCT BEHAVIOUR
Mail provider and queues	Invitations, reminders, confirmations and account messages	Queue delivery asynchronously; expose failures operationally rather than claiming delivery that did not occur.
Swisscom Trust Services	Configured FES/QES identity and signing journey	Preserve one in-flight signing context, correlate callbacks and fail closed when evidence is incomplete.
Payment providers	Paid onboarding and upgrade settlement	Verify amount, currency, payer, order and provider reference server-side; process returns idempotently.
Object storage	Durable source, working and completed document artefacts	Refuse unsafe production fallback when durable storage is unavailable.
AI provider	Optional document analysis and preparation assistance	Report analysis failure explicitly; do not treat AI output as legal approval.
Cardano services	Optional supplementary integrity anchoring and supported wallet payments	Require server-verifiable transaction evidence; never trust browser wallet state as proof.

CUSTOMER ROLLOUT

16. Recommended acceptance checklist

Functional acceptance

- Registration, login, password reset and enabled OAuth providers tested
- Profile and company settings validated with authorised and unauthorised users
- Team invitation, expiry and duplicate-invitation behaviour tested
- German, English, French and Spanish critical journeys reviewed
- Upload, preview, preparation, invitation, signing and download tested
- AES, FES and QES tested for each contracted legal framework

Security and operations

- Tenant and recipient access boundaries tested with direct URLs
- Mail sender, language and tenant links validated from queue workers
- Swisscom success, timeout, cancellation and duplicate callback cases tested
- Payment amount/reference mismatch and duplicate-return cases tested
- Object-storage failure and recovery behaviour tested
- ISO certificate evidence and scope reviewed in the delivery package

Release evidence

Customer production approval should rely on the deployed version, environment-specific configuration, successful end-to-end evidence and the applicable legal/commercial documents—not on this product description alone.

BlockSign GmbH · E-Sign Technical Product Description · V1.0

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Product information only; not legal advice, certification evidence or an SLA.